Quantum Cryptography for IoT Security in Aerospace and Satellite IoT

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Abstract – The adoption of the Internet of Things (IoT) has changed the world of aerospace and satellite communications by facilitating immediate monitoring and data transfer. Unfortunately, these systems remain quite susceptible to cyber-attacks, particularly considering the increasing danger from quantum computing that can shatter classic encryption techniques such as RSA or Elliptic Curve Cryptography. A more secure solution is provided by Quantum Key Distribution (QKD) which is capable of exchanging encryption keys based on the principles of quantum mechanics. This paper investigates two prominent QKD protocols by Bennett and Brassard 1984 (BB84) and Ekert 1991 (E91) and attempts to combine satellite IoT communication with AES encrypted quantum keys. The findings indicate that BB84 provides stronger security with less error and more reliable computational costs than E91, whereas E91 excels in rapid key generation and reduced latency. Security assurance is provided in the form of formal validation showing that both protocols mitigated the risk of cyber threats. The research demonstrates the possibility of implementing quantum cryptographic methods to protect aerospace IoT networks from prospective quantum assaults.

Keywords-- IoT, aerospace, satellite communications, quantum computing, encryption, QKD, BB84, E91, AES, security, formal validation, cyber threats.

I. INTRODUCTION

The aerospace and satellite communication industries have been transformed with the rise of the Internet of Things. It now allows for real-time monitoring, self-governed activities, and efficient data transfer. These developments are vital for satellite-based systems like navigation, meteorology, remote sensing, and military applications. The growth in the number of satellites has opened the door for a multitude of issues, the most problematic being security. These systems can be attacked in many ways, including eavesdropping, data tampering, or gaining unauthorized access. Protecting data during transmission is an obstinate challenge because of the broad range of satellite communication, coupled with satellites' physical inaccessibility. Sensitive data has always been at risk and attempts have been made to protect this data through various means like Rivest-Shamir-Adleman (RSA) and Elliptic Curve Cryptography (ECC), but with the development of quantum computing, these means stand to be highly irrational.

Quantum computers having computational power are able to solve complex mathematical problems at an exponentially higher rate than classical computers. Algorithms such as Shor's algorithm have been discovered to be able to break RSA and ECC encryption, thus making traditional security measures more and more obsolete. The aerospace industry and the satellite IoT sector are therefore forced to employ more advanced cryptographic protocols that are quantum-resistant. Quantum Key Distribution (QKD) offers a viable solution by using principles of quantum mechanics to create secure encryption keys that are resistant to interception or copying without being detected. Unlike traditional cryptographic key exchange protocols, QKD causes any attempt at eavesdropping to alter the quantum state of the information being transmitted, which would immediately alert the parties to the communication.

This study investigates the incorporation of Quantum Key Distribution (QKD) in satellite Internet of Things (IoT) security frameworks using and comparing two well-known QKD protocols—Bennett-Brassard 1984 (BB84) and Ekert 1991 (E91). The keys obtained from the two protocols are used in combination with Advanced Encryption Standard (AES) encryption for enhancing data security in satellite-based IoT communications. The performance of the protocols is evaluated on key parameters of encryption speed,

latency, compute overhead, and security robustness using simulation with Qiskit. Security testing by simulation of adversarial cyber-attacks like eavesdropping, man-in-the-middle attacks, and ciphertext modification attacks is performed to test quantum-encrypted resilience.

By integrating quantum cryptography with standard encryption methods, this work hopes to create an extremely secure yet scalable solution to safeguard aerospace and satellite IoT networks. The findings offer vital perspective on the practicability of implementing quantum-secured communication within real-world systems, guaranteeing confidentiality and data integrity against foreseeable future quantum computer attacks. With ongoing advancement of quantum technology, the implementation of QKD-based security measures will be imperative to safeguard sensitive satellite communications against potential future cyber-attacks.

II. LITERATURE REVIEW

[1] Research undertaken has focused on the BB84 protocol as well as DV-QKD to increase the overall efficiency, but the high overhead associated with communication during key sifting, error correction, and verification remains a challenge. A traditional method uses excessive bit exchanges which renders the approaches as not very efficient. This document is using three PRNG-based optimizations: comprehensive chunked bit positioning that reduces sifting overhead, hash-based key verification that makes use of lightweight key checking, and adaptive basis reconciliation that lowers communication costs. These advances reduce overhead by 50%, making them considerably easier to implement in other QKD protocols and making the use of quantum communication for practical purposes more feasible.

[2] Previous work on IoT enabled ride sharing has used classical cryptography to address the security and privacy issues but those are vulnerable to quantum attacks and not long term secure. Threshold Private Set Intersection (TPSI) has been explored as a privacy preserving approach but existing TPSI protocols are not quantum secure. This work introduces the first QC based TPSI protocol using quantum cryptography principles like Heisenberg uncertainty principle and no-cloning theorem to be secure against quantum attacks. By adding QC the proposed solution provides privacy in ride sharing and long term security which is very much relevant for secure IoT applications including aerospace and satellite IoT.

[3] Previous research on NB-IoT security has focused on traditional cryptographic methods which are vulnerable to quantum attacks and require complex key management systems like PKI. Position driven cryptography solutions also have security limitations like collusion attacks and location spoofing. This paper introduces a new location aware cryptographic system using lattice based cryptography, providing post-quantum security for IoT. Unlike previous methods it does not require PKI, authentication without SIM cards and is resistant to quantum attacks and location spoofing. This makes it very suitable for secure IoT applications like aerospace and satellite IoT where secure and efficient communication is critical.

[4] Previous work on secure satellite image transmission has looked at various encryption techniques and chaos based encryption has emerged as a promising one due to its high security, low computational overhead and ease of implementation. Existing works have analyzed chaotic systems for image encryption and evaluated their performance and security under different scenarios. However, challenges still remain such as vulnerability to certain attacks and need for better performance in resource constrained environment like satellite IoT. This paper provides a comprehensive review of chaos based encryption for satellite communication, its advantages and limitations and open research problems. Its findings are relevant to aerospace and satellite IoT security where robust and efficient encryption is crucial for protecting image data.

[5] Previous research has shown that cyber attacks and signal jamming on space based assets which are critical to global infrastructure are on the rise. Traditional cybersecurity is struggling to protect these systems against non-kinetic attacks. Quantum technologies offer both offensive and defensive capabilities to counter cyber threats from state and non-state actors. While the big powers are investing in quantum based security, the field is still in its infancy and practical applications are limited. This paper argues for the need for advanced quantum cryptographic solutions to protect aerospace and satellite IoT systems from future cyber threats.

[6] Previous research on SATCOM security has looked into physical layer security and cryptographic methods to counter the evolving cyber threats. While advances in radio technology and manufacturing has improved connectivity and reduced cost, it has also increased the attack surface making SATCOM systems more vulnerable. Existing studies categorize security approach into anti-jamming, anti-spoofing and quantum based key distribution and highlights their pros and cons. However industry driven deployments often prioritizes performance over security and thus trade-offs are made that exposes vulnerabilities. This paper emphasizes the need for robust quantum cryptographic solutions to improve SATCOM security especially for aerospace and satellite IoT applications where secure and resilient communication is key.

[7] Previous research has shown security risks in UAV communications due to their open wireless channels and limited resources. Traditional cryptography is vulnerable to quantum attacks as Shor's and Grover's algorithms can break current encryption schemes. While PQC is being developed and standardized by NIST, challenges

remain to implement these algorithms on resource-constrained UAVs. This paper reviews UAV security threats, PQC development and future challenges and highlights the need for quantum resistant cryptographic solutions to protect aerospace and satellite IoT systems from emerging threats.
[8] Previous research has shown the security challenges in UAV Traffic Management (UTM) where the communication links between UAVs, pilots and UTM systems must meet strict safety and integrity requirements. Traditional internet cryptographic standards are not suitable due to UAVs limited resources and dynamic environment. Research has explored authentication protocols, alternatives to AES and securing UTM protocols like ADS-B and RemoteID. However existing solutions lack a comprehensive security architecture that ensures authentication and data integrity across all communication links. This paper emphasizes the need for quantum resistant cryptographic frameworks to enhance UAV security especially in aerospace and satellite IoT applications.

[9] Previous research on satellite authentication has used traditional cryptographic methods which have issues in space due to key management and quantum attacks. This work proposes a quantum-safe authentication scheme using quantum entanglement and measurement noise profiles to create unique "quantum noise fingerprints" for satellite nodes. The method consists of a training phase to establish noise profiles and an online authentication phase for real-time verification. By using quantum mechanics and system imperfections this makes it more secure and reliable for satellite constellations. We demonstrate our concept using IBM System One, and it can be used for secure aerospace and satellite IoT communications.

[10] Previous research has shown that ground-based satellite operations are vulnerable to cybersecurity threats. Traditional IP cryptography is vulnerable to quantum attacks as quantum computing advances. This research aims to develop a quantum-resistant cryptographic framework to secure data transmission between ground stations and satellites. By finding a resilient protocol that can withstand quantum attacks, it will secure IP-satellite communications and ensure reliable and secure operations in aerospace and satellite IoT networks.

[11] Previous research has shown that satellite communication is under increasing cyber-attacks which is critical for industries like telecom and navigation. Traditional security frameworks rely on encryption, authentication and intrusion detection to mitigate risks. This research proposes a multi-layered security approach, analysing vulnerabilities and integrating various protective measures to ensure data confidentiality, integrity and availability. Through simulations and testing, the approach has been proven to mitigate cyber threats. But with the advent of quantum computing, further quantum-resistant encryption is needed to secure aerospace and satellite IoT networks from future cyber threats.

[12] With cyber threats in aerospace growing, previous research has looked at advanced defence mechanisms, AI, blockchain, quantum cryptography and biomimicry-inspired architectures. AI driven anomaly detection, reinforcement learning and generative adversarial networks (GANs) are key to real time threat identification and mitigation. Blockchain provides tamper proof logging while resilient system architectures provide adaptability. Quantum cryptography, particularly Quantum Key Distribution (QKD) is highlighted as the solution to secure satellite communications against quantum attacks. While these technologies offer solutions, challenges remain in standardisation, global collaboration and real world implementation. More research is needed to refine quantum safe cryptographic frameworks for aerospace and satellite IoT security.

[13] With the advent of increasing digital aviation communications, previous research has focused on developing quantum-secure protocols to respond to the security challenges in aircraft and ground station communications. This study introduces two hybrid key exchange protocols, PQAG-KEM and PQAG-SIG, using post-quantum and classical cryptographic mechanisms, which provide protection against existing and future threats by quantum computers. The PQAG-KEM keeps the key exchange secure, while PQAG-SIG provides authentication based on quantum-safe signatures. Implementation and assessment of the protocols are done against the backdrop of existing secure aviation protocols, demonstrating better security and efficiency. However, challenges are met in terms of the adaptation of these protocols to the strict upgrade constraints of aviation, as well as in ensuring that they can coexist seamlessly with existing systems.

[14] Prior research has disclosed the extensive examination of the impact of quantum computing on cryptography, particularly in conducting cryptanalyses of classical encryption methods such as RSA and ECC. This study proceeds to outline such quantum properties as superposition and entanglement that enabled, among other feats, quantum computation and developments in cryptography. A major theme of this discussion is on quantum encryption techniques, QKD, post-quantum cryptographic algorithms, which promise to provide resilience against quantum threats. These solutions show promise in securing future communications; however, large-scale deployment and integration into extant aviation and IoT security systems remain burdensome.

[15] This mission is a prime accomplishment for space-borne quantum key distribution (QKD) paving the way for the first sovereign QKD system in Europe-accompanied by funding from ESA and SES, alongside several national agencies and private

partners-with a view of developing the quantum technology based secure communication network. Key developments include a QKD transmitter onboard a Low Earth Orbit (LEO) satellite and a modification of the Optical Ground Station Oberpfaffenhofen (OGS-OP) for in-orbit testing. The mission contains studies on adaptive optics to aid in quantum signal reception and other challenges like atmospheric distortions. Although this is a big leap toward secure aerospace communications, roadblocks toward practical application and large-scale deployment stand in the way of future quantum secure Internet of Things networks in satellite communications.

III. PROPOSED SYSTEM

A. System Architecture

The system is designed to secure satellite IoT networks by combining Quantum Key Distribution (QKD) with AES encryption. Traditional encryption methods are vulnerable to quantum attacks so we use BB84 and E91 QKD protocols to generate cryptographic keys for encryption.

The architecture consists of three parts:

- **Quantum Key Distribution (QKD):** Secure key exchange using quantum mechanics, no eavesdropping.
- **AES Encryption:** Use quantum generated keys to encrypt satellite data for confidentiality.
- **IoT Transmission Model:** Secure transfer of encrypted data over satellite networks using socket programming.

By combining quantum cryptography with AES encryption we have a scalable and future proof solution for satellite IoT.

B. Integration of QKD with AES

Integration of two quantum key distribution protocols: BB84 and E91, in this system, contribute to developing a robust mechanism for key exchange. These protocols allow for the generation and sharing of encryption keys, such that any endeavour at interception or manipulation by an adversary would be detectable.

Integration characteristics include:

- **BB84 and E91 Protocols:** Use of protocols to generate quantum-resistance encryption keys.
- **Secure Storage and Exchange of Keys:** Guarantees secured storage of the generated keys and transmits them for use in AES encryption.
- **Real-time Key Distribution:** is achieved using socket programming in order to provide secure and effective communication between the sending and receiving nodes.

The combination of QKD with AES encryption ensures a proper balance between quantum security and computational efficiency, strong encryption and practical implementation.

C. IoT Satellite Communication Model

The system transmits data across **satellite IoT networks** securely, addressing **eavesdropping, unauthorized access and data tampering**. The communication pipeline is as follows:

1. **Data Collection:** IoT satellite sensors collect real-time data (e.g. navigation, meteorology, remote sensing).
2. **Quantum Key Exchange:** Secure keys are distributed using **BB84 and E91 protocols**.
3. **AES Encryption:** Collected data is encrypted using AES with the quantum keys.
4. **Secure Transmission:** Encrypted data is sent via **socket programming**, confidential and secure from cyber threats.
5. **Performance & Security Evaluation:** The system is tested for encryption speed, latency and resistance to attacks like **eavesdropping, man-in-the-middle and ciphertext modification** using Qiskit simulations.

By putting **quantum-secured encryption** into real satellite IoT applications, this model secures the data and keeps the communication efficient. With **quantum technology** advancing rapidly, **QKD based security solutions are a must to protect satellite communications from future cyber-attacks.**

IV. METHODOLOGY

A. Quantum Key Distribution (QKD) Implementation

The study realizes two QKD protocols-BB84 and E91-for secure key establishment between sender and receiver. These protocols create a shared secret key and enforce minimum security against eavesdropping.

a. BB84 Protocol

- The sender generates a string of independent random qubits in either rectilinear or diagonal basis and sends them to the receiver.
- The receiver measures the qubit received with random bases.
- Both parties correct for mismatched bases and discard all misaligned measurements.
- A portion of the key is used to estimate the QBER.
- After error correction and privacy amplification, the final secret key is distilled.

b. E91 Protocol

- The sender and receiver share entangled photon pairs generated using Bell states.
- Each of the parties measures the entangled qubits with randomly chosen bases.
- The measurement results are publicly compared to see whether the entangled correlations are maintained.
- Extraction of a joint secret key is carried out following the secure-don't-be-a-thing-opposite assumption.

B. AES-Based Secure Communication

After the secret key has been generated via the QKD protocols, it will be used to encrypt and decrypt images using the AES algorithm.

a. Encryption Process

- The sender employs AES, which is a symmetric encryption scheme, to encrypt the image, with the QKD-generated key acting as the encryption key.
- The encrypted image is sent to the receiver via socket programming over a communication channel.

b. Decryption Process

- The receiver uses the shared QKD key to decrypt the received image using AES.
- The original image is reconstructed upon successful decryption.

C. Socket Programming for Secure Transmission

Socket programming is one way of developing real-time communication between the sender and receiver:

- The sender initiates a connection to the receiver.
- Sending of encrypted image data through the socket is done.
- The receiver, which is waiting to receive data, does end up receiving data and then decrypts it.

D. Security Validation and Attack Simulation

To evaluate the effectiveness of the QKD-AES scheme, we simulate different attack scenarios:

- **Eavesdropping Attack:** We simulate this by using an intercept-resend method to measure changes in the Quantum Bit Error Rate (QBER).
- **Man-in-the-Middle Attack:** In this scenario, an attacker tries to modify the messages exchanged during the key exchange process.
- **Ciphertext Tampering Attack**: We examine the impact of modifying the encrypted quantum key or transmitted data to observe how it affects decryption and system security.

E. Performance Metrics Evaluation

Security and effectiveness of the proposed system are assessed using the following performance measures:

- **Quantum Bit Error Rate (QBER):** The error rate of the received key as a result of eavesdropping or noise.
- **Entropy and Randomness Analysis:** Guarantees that the produced key possesses actual randomness and unpredictability.
- **Key Agreement Ratio:** Quantifies the trustworthiness of exchanged keys.
- **Encryption and Decryption Speed:** Regulates computational performance.
- **Latency and Overhead:** Measures the time required for key generation, encryption, transmission, and decryption processes.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Simulation Setup

a. Tools and Platforms Used

- **Qiskit (IBM Quantum)**: Simulated quantum circuits and key exchange.
- **Python**: Implemented key processing, attack simulations, and performance analysis.
- **Cryptography Library**: Used for AES encryption and security validation.
- **Matplotlib & NumPy**: Utilized for visualization and entropy/randomness calculations.

b. Experimental Conditions and Test Scenarios

- **Quantum Key Distribution (QKD) Protocols**: BB84 and E91 key exchange mechanisms were simulated.
- **Key Length**: Randomly generated key lengths for each trial.
- **Attack Scenarios**: Eavesdropping, Man-in-the-Middle (MITM), and Ciphertext Tampering.
- **Performance Metrics**: Key Generation Rate (KGR), Quantum Bit Error Rate (QBER), Shannon entropy, key agreement ratio, encryption speed, latency, and computational overhead.
- **Hardware Specifications**: Simulated on a laptop with an Intel i7 processor and 16GB RAM.

B. Performance Comparison: BB84 vs. E91

TABLE 1. KEY EXCHANGE EFFICIENCY

Metric	**BB84**	**E91**
Key Generation Rate (KGR)	76,260.07 bits/sec	500,812.42 bits/sec
Key Agreement Ratio	0.8000%	240.00%
Total Time Taken	0.0001 seconds	0.0000 seconds

a. Computational Complexity Analysis

- **BB84**: Requires classical post-processing and measurement-based qubit transmission, leading to additional processing time.
- **E91**: Uses entanglement-based key distribution, resulting in faster key generation and higher security.

C. Security Evaluation

TABLE 2. QUANTUM BIT ERROR RATE (QBER) ANALYSIS

Attack Scenario	**BB84 QBER**	**E91 QBER**
Without Attack	0.00%	0.00%
Eavesdropping Attack	40.00%	0.00%
Man-in-the-Middle Attack	20.00%	0.00%
Ciphertext Tampering Attack	20.00%	0.00%

TABLE 3. ENTROPY AND RANDOMNESS ANALYSIS

Scenario	**BB84 Entropy (bits)**	**E91 Entropy (bits)**

Without Attack	1.0000	0.9183
Eavesdropping Attack	0.7219	0.8813
Man-in-the-Middle Attack	0.7219	0.9710
Ciphertext Tampering Attack	0.7219	0.9710

- **BB84 QBER is highly impacted by attacks**, especially eavesdropping.
- **E91 demonstrates greater attack resistance**, showing stable entropy and QBER across all scenarios.

D. Resource Utilization and Optimization Results

TABLE 4. ENCRYPTION PERFORMANCE METRICS

Metric	**BB84**	**E91**
File Size	4.83 MB	4.83 MB
Encryption Speed	0.014338 sec	0.018469 sec
Encryption Latency	0.000880 sec	0.000177 sec
Computational Overhead	0.011913 sec	0.019808 sec

a. CPU and Memory Usage Comparison

- **BB84**: Moderate CPU usage due to classical post-processing.
- **E91**: More efficient in key generation, but slightly higher encryption overhead.

b. Trade-off Between Security and Efficiency

- **BB84 is simpler to implement**, but is more vulnerable to attacks.
- **E91 offers stronger security and faster key generation**, but has a slightly higher computational cost in encryption.
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VI. DISCUSSION

A. Implications of Quantum Cryptography for IoT Security

The combination of QKD with AES encryption in space-based IoT communication networks is promising to solve some post-quantum security challenges. Unlike classical encryption schemes, which lie defenceless against Shor's algorithm and other quantum-based attacks, QKD offers unconditional security through the fundamental laws of quantum mechanics.

a. Real-World Applicability in Aerospace and Satellite Networks

As per this research analysis, it was concluded that the proposed QKD-AES model can significantly enhance security in satellite-IoT applications, including the following:

- **Navigation Systems** – Ensuring integrity and confidentiality in GPS and GNSS communications.
- **Military Communications** - Secure transmission of classified data over satellite channels.
- **Meteorology and remote sensing** - Unauthorized access/data manipulation.

Even though BB84 and E91 QKD protocols differ in strength, their integration with AES encryption provides protection to data confidentiality and helps resist quantum attacks in space-based IoT networks.

b. Advantages Over Classical Cryptographic Methods

QKD offers the following advantages over conventional cryptographic methods such as RSA and ECC:

- **Future-Proof Security**: Attacks from large quantum computers are resistant.
- **Quantum Safe Key Distribution:** Detection of eavesdropping allows for the deterrent reward of shipping keys without apparently intercepting them.
- **High Entropy and Randomness:** Keys generated by quantum mechanics are more entropic and random, reducing the odds of guessing one.

B. Challenges and Limitations

Notwithstanding these significant security guarantees, real-world IoT network implementation of QKD would encounter several challenges:

a. Implementation Complexity

- **Classical Post-Processing Overuse:** BB84 needs additional steps for error correction and privacy amplification, leading to a higher computational price.
- **Synchronization Issue:** E91 needs entanglement, which complicates the synchronization problem between the transmitter and receiver.

b. Hardware Requirements for Real-World QKD Deployment

- **Quantum Channel Infrastructure:** Deploying QKD requires separate optical fibres or free-space optics, rendering the process challenging for large-scale application in satellite communications.
- **Resource Constraints:** Computational and memory requirements of QKD systems may be beyond the capacity of practical IoT devices.
- **Noise and Error Sensitivity:** Variables in the environment, like atmospheric interference, will have a tendency to augment the Quantum Bit Error Rate (QBER), thus influencing key generation efficacy.

C. Future Research Directions

To overcome these limitations, future research should focus on optimizing quantum cryptography for real-world deployment**.**

a. Potential Advancements in Quantum Cryptography

- **Hybrid QKD models:** This has given rise to models like BB84 plus E91 that combine classical encryption protocols to bring a balance of security and efficiency**.**
- **Error Mitigation Techniques:** Adaptive error mitigation algorithms are proposed to help manage QBER and thus improve the rates of key agreement.
- **Hardware Efficient QKD Implementations:** A path towards quantum hardware with reduced resource requirements.

b. Integration with Post-Quantum Cryptographic Methods

While QKD provides a high level of security assurance, its deployment challenges warrant hybrid approaches which may combine QKD with post-quantum cryptography (PQC) algorithms like:

- Lattice-Based Cryptography
- Code-Based Cryptography
- Multivariate Polynomial Cryptography

QKD applications blending PQC pave the way toward a scalable, feasible, and quantum-resilient security framework for future satellite and IoT communication systems.

VII. CONCLUSION

A. Summary

The integration of QKD with AES encryption provides enhanced security in satellite-based Internet of Things communication networks. Two QKD protocols, BB84 and E91, were analysed and compared based on key exchange efficiency, security resilience, and computational overhead. The analysis proposes:

- BB84 has thus far been secure, with low rates of error and easier software and hardware integration, but is open to eavesdropping attacks.
- E91, faster in generating keys, better in key agreement rates, and attacks-resilient, but demanding in terms of an encryption's computational overhead.
- AES encryption with quantum security protects satellite IoT data against cyber threats, assuring confidentiality and integrity.
- Attack simulations including eavesdropping, man-in-the-middle, and ciphertext tampering attacks confirmed the fact that QKD-AES encryption is robust against quantum adversaries.

B. Contributions of the Study

The contributions of this research to quantum cryptography and the security of IoT can be summarized as follows:

- Demonstration of the feasibility of a QKD-based key exchange, combined with AES encryption, for satellite IoT networks.
- Security performance, computational efficiency, and attack resistance of BB84 and E91 protocols were evaluated and compared.
- Secure IoT satellite communication model is implemented with socket programming and provides real-time data encryption and transmission.
- Simulated quantum-based attacks to validate the security resilience of QKD-encrypted communication systems.

C. Final Remarks

As quantum computing develops rapidly, older cryptographic techniques will increasingly fall prey. This research showcases QKD as a promising solution for the protection of satellite-based IoT networks against quantum threats. To some extent, hardware complexity, implementation costs, and routing path constraints are already there-but innovations in practical quantum hardware and post-quantum cryptographic methods create the conditions for resistance to QKD security techniques to flourish.

The fusion of quantum cryptography with classical encryption can assure a future-proof measure against aerospace and satellite communication security. The benefits of evolutionary development in quantum technologies will in future call for the adoption of QKD-enhanced security frameworks for the protection of critical IoT infrastructure against emerging cyber threats.

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